

DATA PAPER

Urban vegetation data across 12 cities in northeastern North America

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Handling Editor: Miguel A. Acevedo**Abstract**

Datasets describing the composition of biological communities across broad anthropogenic gradients are valuable resources for ecologists and conservation biologists, with applications ranging from testing theories of community assembly to guiding biodiversity management. But much of the most rigorously collected field survey datasets, particularly for vascular plant communities in urban environments, remain underutilized because they are not curated or publicly accessible. We compiled and standardized a large database of information on the composition of vascular plant communities from sites spanning urbanization gradients in 12 cities in northeastern North America. The database integrates 26 field studies conducted by researchers from 1987 to 2024, encompassing 746 sites of either forest ($n = 408$ sites) or open habitat (e.g., meadow, prairie, vacant lot; $n = 338$ sites). Forest sites include canopy (overstorey) and ground (understorey) layers, while open habitats only include a ground layer. Field sites span a gradient of urbanization from rural to urban areas and are characterized as being minimally managed in situ at the time of

For affiliations refer to page 2

surveys. Thus, the data are representative of spontaneous vegetation that was not seeded or planted. In total, the database contains records for 1296 taxa identified to the species level, including 322,101 presence/absence records and 23,805 abundance records, all geo-referenced and accompanied by metadata on site area, sampling effort, site management, site history, and methodology, all of which varied by site. To ensure consistency across studies, taxonomic names were standardized according to the Database of Vascular Plants of Canada (VASCAN), and all measurements were summarized at the site level. Unlike presence-only data commonly available from citizen science repositories, this database provides both presence/absence and abundance records collected through systematic surveys. As a result, it avoids many of the pitfalls associated with citizen science data such as opportunistic sampling, bias towards public/highly frequented sites, lack of a complete record of all species in a community, and misidentifications by nonexpert observers. It also enables analyses that would otherwise be difficult or impossible, such as studies of species abundance, community composition, rare taxa, and species of low desirability. Overall, this database offers a valuable resource for those interested in studying plant community dynamics and the impacts of urbanization and other anthropogenic gradients on biodiversity. The data are available for reuse under CC0 1.0 Universal licensing; however, we appreciate the citation of this data paper when reusing the dataset.

KEYWORDS

abundance, community composition, community ecology, field survey, North America, occurrence, presence/absence, urban gradient, urbanization, vascular plants

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data are available in Data S1. These data are also available in Dryad at <https://doi.org/10.5061/dryad.b2rbnzsx5>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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